Do Modern Time-Series Explainers Transfer to the Cloud? An Empirical Study on Workload and Network Traffic Forecasting

Mohamed Readh FENTAZI
Communication Systems Department
EURECOM
Sophia Antipolis, France
fentazi@eurecom.fr

Adlen KSENTINI
Communication Systems Department
EURECOM
Sophia Antipolis, France
ksentini@eurecom.fr

***Abstract*—Deep learning models are increasingly used to forecast workloads, network traffic, and resource utilization in cloud platforms and data centers, where predictions drive autoscaling, capacity planning, and orchestration decisions. Yet their black-box nature limits operator trust and hinders adoption in production environments. Explainability practice in cloud and network management still relies largely on general-purpose techniques such as SHAP and LIME, which were not designed for temporal data and ignore dependencies across time steps. Meanwhile, the time-series XAI community has produced a new generation of specialized methods that have yet to be applied to operational cloud data, including learned perturbation masks, temporal Shapley variants, time-series counterfactuals, and frequency-domain attribution. In this paper, we present an empirical study of these state-of-the-art time-series XAI methods in the context of cloud workload and data-center network traffic forecasting. Using publicly available cluster and network traces and several deep forecasting architectures, we evaluate the resulting explanations in terms of faithfulness, robustness, sparsity, and computational cost, and examine how cloud-specific characteristics such as burstiness, flash crowds, strong periodicity, and non-stationarity affect each family of methods. Our study assesses whether the state of the art in time-series explainability transfers to cloud networking, and aims to provide practical guidance for building explainable forecasting pipelines for cloud resource management and orchestration.**

***Index Terms*—explainable AI, time series forecasting, cloud computing, workload prediction, data center networks, resource management, deep learning**